Consumer Valuation of Battery Health Information in Used Electric Vehicle Markets: A Discrete Choice Experiment

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ABSTRACT

The used battery electric vehicle (BEV) market is growing rapidly, yet consumers face substantial uncertainty about battery condition at the point of sale. This paper estimates consumer preferences and willingness to pay (WTP) for battery health information in the used BEV market using a discrete choice experiment (DCE) administered to a national sample of 2,916 U.S. adults. The DCE jointly varies conventional vehicle attributes (price, mileage) and battery-specific attributes (electric range, degradation rate, refurbishment history) across six repeated choice tasks. We estimate a mixed logit model (MXL) and a six-class latent class choice model (LCCM) to capture both aggregate preferences and preference heterogeneity. At the population level, consumers value vehicles with lower mileage andadditional range, while penalizing range degradation and battery refurbishment. The LCCM reveals six distinct consumer segments that differ markedly in their sensitivity to battery quality signals, price, and the no-choice option. Approximately 62% of the sample belongs to classes that assign significant disutility to refurbishment history, suggesting substantial demand for credible battery health disclosure at the point of sale. An information treatment analysis further shows that providing detailed battery and warranty information reshapes the type of concern driving opt-out behavior, but does not reduce overall market disengagement. These findings carry direct implications for used vehicle dealers, OEMs, and certification service providers seeking to develop market infrastructure for used BEVs.

1. Introduction

The used vehicle market plays a central role in U.S. automobile ownership: used vehicle transactions now account for roughly 70% of all passenger vehicle purchases and are projected to grow steadily through the coming decade (Market Research Future, 2025). As electric vehicles (EVs) continue to penetrate the new car market, an increasing number of BEVs are entering the secondary market. Used EV sales rose 34.2% year-over-year through early 2025, while average days of supply fell by 21.5%, signaling rapid and accelerating demand (Cox Automotive Inc., 2025). The growth of the used BEV market extends EV access to middle- and lower-income households (Hagman et al., 2016), supports a circular economy by extending vehicle and battery lifetimes (Guzek et al., 2024), and underpins the residual values that make new EV leases financially viable (Brückmann et al., 2021; Lim et al., 2015).

Despite this momentum, consumer hesitation toward used BEVs remains pronounced. Fernandes (2023) indicates that 52% of U.S. adults are unlikely to consider purchasing a used EV, compared to only 28% who express openness to the idea. Consumers frequently cite uncertainty about vehicle and battery condition (41%) and concerns about reliability relative to new EVs (23%) as primary barriers. Among those who have already purchased a used EV, over half reported pre-purchase concerns about battery longevity, yeteven though only 5.1% experienced substantial performance loss in practice (Sheykhfard et al., 2025). This gap between perceived and actual risk points to a fundamental information asymmetry: buyers lack information and knowledge about the battery at the point of sale, which represents up to 30% of a used BEV's value (Boudway, 2020).

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Unlike conventional vehicles, where odometer reading and vehicle age serve as reasonable proxies for overall wear, battery degradation in BEVs depends on factors that are invisible to buyers: charging history, depth of discharge cycles, thermal exposure, and driving patterns (Bashash et al., 2011; Neubauer et al., 2012). Two BEVs of identical age and mileage can differ substantially in their remaining battery capacity and projected useful life. This creates a classic lemons problem in which sellers of well-maintained vehicles cannot credibly signal battery quality, depressing prices across the board and discouraging investment in battery upkeep. The consequences extend beyond individual transactions: persistent market uncertainty may suppress new EV adoption by generating "resale anxiety" among prospective buyers who question whether their new vehicle will hold its value (Brückmann et al., 2021).

This paper asks how consumers value battery health information in the used BEV market and how preferences vary across consumer segments. We address these questions using a discrete choice experiment (DCE) administered to a national U.S. sample of 2,916 adults who reported an intention to purchase a used car or SUV within the next two years. The DCE presents respondents with repeated choices among hypothetical used BEVs that vary in mileage, purchase price, electric range, battery degradation rate, and battery refurbishment history. Respondents are also randomly assigned to receive either basic range-and-health information or extended information that additionally describes battery maintenance costs and warranty coverage, allowing us to assess whether information treatment shapes preferences and opt-out behavior.

We estimate two complementary models. A mixed logit (MXL) model in WTP space characterizes aggregate preferences and the degree of preference heterogeneity across the full sample. A six-class latent class choice model (LCCM) then identifies discrete consumer segments that differ in their sensitivity to price, battery quality signals, and the outside option. To our knowledge, this study provides the first WTP estimates for battery health information in the U.S. used BEV market.

The MXL results confirm that the population, on average, values additional electric range ($11,160 per 100 miles at Year 3), penalizes higher mileage ($2,810 per 10,000 miles), and discounts battery degradation rate ($1,000 per percentage point of annual range loss). Both refurbishment types carry significant disutility, though large standard deviation estimates indicate substantial variation across individuals. The LCCM reveals that this heterogeneity is structurally organized: approximately 62% of respondents belong to three classes that strongly penalize any refurbishment history. A further 17% are either price-constrained buyers largely indifferent to battery quality signals or deeply skeptical non-adopters unlikely to enter the used BEV market in the near term. The remaining class consists of highly committed, price-insensitive buyers who place an exceptionally high premium on a pristine battery service history. The information treatment analysis reveals that extended battery information raises the salience of battery-specific concerns among opt-outs but does not reduce the overall rate of market disengagement.

These findings are relevant to dealers, OEMs, battery service providers, and financing institutions that are developing commercial infrastructure for used BEVs. The existence of large, identifiable segments with strong preferences for battery quality documentation supports the viability of premium pricing for certified battery health, while the presence of a price-sensitive but engaged segment points to the importance of affordable product offerings. We discuss the business and market implications of these findings in detail in Section **??**.

The remainder of the paper is organized as follows. Section 2 reviews the relevant literature. Section 3 describes the survey design and sample. Section 4 presents the econometric framework. Section 5 reports results. Section **??** discusses market and business implications. Section 6 concludes.

2. Literature Review

This section reviews key literature informing survey design and analyses of this study. The first subsection examines the main drivers and barriers to the adoption of used BEVs, while the second subsection focuses on EV battery technology and how it shapes consumer perceptions and decision-making.

2.1. Used Electric Vehicle Adoption

Gore et al. (2024) provided a comprehensive review on consumer perspectives on EVs and the factors influencing EV adoption, identifying a broad range of determinants including attitudinal, demographic, societal, environmental, technological, and economic variables (Iogansen et al., 2023; Naseri et al., 2024; Sonar et al., 2023). However, most existing studies focused on new EV markets, with comparatively limited attention given to the used EV markets, which is rapidly expanding and playing an increasingly important role in the broader diffusion of electric mobility.

Though some preferences may carry over between new and used vehicle buyers—for instance, longer driving range consistently correlates with stronger EV preference—emerging evidence suggests that these two consumer groups differ in meaningful ways. For instance, used EV buyers are more sensitive to the availability of public charging infrastructure than new EV buyers, likely due to lower rates of home charging access (Zou et al., 2020). Moreover, purchasing behavior for used vehicles is often more risk-averse and budget-conscious, reflecting different decision-making processes than those found in the new car market. This indicates that used EV adoption is not simply a scaled-down version of new EV adoption and warrants independent investigation.

Several studies have begun to profile the typical used EV buyer. Canepa et al. (2019) found that used car buyers are commonly cost-conscious. Compared to the general population, used EV buyers are more likely to be younger, male, White, and better educated. However, compared to new-EV buyers, used EV buyers tend to have lower household incomes and be renters, with lower access to private garages with EV chargers (Khaloei et al., 2020; Sheykhfard et al., 2025). These characteristics differ from the traditional early adopter profile seen in the new EV market (typically wealthier homeowners with garages), suggesting that used EVs may serve as a more accessible entry point into EV ownership for broader population segments.

Although range is a common concern for EV users, Sheykhfard et al. (2025) found that the majority of used EV owners drive well within the range capabilities of most existing EV models: 59.4% reported driving fewer than 100 miles per day and 29.4% drove less than 50 miles per day. This implies that range anxiety may be less of a practical barrier for used EV buyers, who tend to have shorter commutes or use their vehicles in more limited contexts. Range may therefore be a less decisive factor in used EV adoption compared to other attributes such as battery condition, charging accessibility, and price.

Electric vehicle batteries (EVBs) present new risks and opportunities for both consumers and the automotive industry. Despite progress in battery technology, consumer concerns about battery health, replacement cost, and resale value persist, particularly in the used EV market. Battery costs can represent up to 30% of a vehicle's total price (Boudway, 2020), making battery longevity and reliability central to the economic attractiveness of used EV ownership.

2.2. Battery Degradation and State-of-Health

In the EV adoption literature, battery-related considerations have largely been operationalized through driving range (Yuan et al., 2018). Prospective used EV buyers commonly express concerns about battery degradation, SOH uncertainty, and the perceived unreliability of older battery technology (Pedrosa and Nobre, 2018). Battery degradation manifests as a reduction in energy capacity, driving range, and charging efficiency over time (Attia et al., 2022). The rate of degradation depends on charging patterns, driving behavior, depth of discharge, and operating conditions such as temperature exposure (Bashash et al., 2011; Neubauer et al., 2012). Yang et al. (2018) estimates that BEV batteries can last between five and thirteen years depending on climate conditions. Battery aging trajectories tend to be linear or sublinear, as degradation rates often stabilize after initial use due to the settling of battery chemistry (Attia et al., 2022, 2020). A battery is generally considered to have reached its End-of-Life (EoL) for vehicle use when its SOH drops to approximately 70–80% of original capacity (Canals Casals et al., 2022, 2019). Battery degradation reduces the perceived future value of a used BEV and heightens resale anxiety among prospective buyers.

2.3. Battery Replacement, Cost, and Warranty Considerations

While the literature has examined battery longevity in the context of total cost of EV ownership (Hagman et al., 2016; Letmathe and Suares, 2017), direct evidence on the cost of battery replacement in used BEVs remains limited. Most modern EVBs are designed to outlast the vehicle's useful life under normal use. Dnistran (2024), in a study of nearly 5,000 EVs, estimated an average annual degradation rate of just 1.8%. Nevertheless, when replacement is required, costs can range from \$5,000 to \$20,000 or more depending on vehicle model and labor (Kothari, 2024; Witt, 2024). To support consumer confidence, most EV manufacturers provide warranties covering 8 years or 100,000 miles, whichever comes first (Clarke, 2024). For used vehicle buyers evaluating whether a battery's SOH will remain above the 70–75% warranty threshold within their ownership window, the presence of remaining warranty coverage represents meaningful additional value. Letmathe and Suares (2017) further account for battery residual value as a separate component of total cost of ownership, underscoring the economic significance of battery condition at resale.

Second-Life and Refurbished Batteries As the number of retired EVBs increases, second-life applications and refurbishment are gaining commercial attention (Skeete et al., 2020; Tankou et al., 2023). The two dominant second-life pathways are repurposing degraded batteries for stationary energy storage and refurbishing them for continued use in EVs. This paper focuses on the latter. Refurbishing batteries at the cell or module level can lower vehicle costs and support the commercial viability of BEVs: Jiao and Evans (2016) show that repurposing EoL EVBs could reduce the cost gap between BEVs and conventional vehicles, while Shaikh et al. (2023) demonstrate through simulation that refurbished EVBs are less expensive than new ones, with cost advantages that are greatest in local markets. However, existing research on second-life EVBs has focused primarily on technical and economic feasibility, with comparatively little attention to consumer demand for refurbished options in the U.S. context.

One of the few consumer-facing studies, conducted in Portugal, found that respondents expressed positive attitudes toward used EVs fitted with replacement batteries backed by dealership warranties, and indicated higher WTP relative to vehicles with original batteries (Pedrosa and Nobre, 2018). This qualitative study was small in scope and may not generalize to the U.S. market. More importantly, no study to date has examined consumer valuations of partial versus full battery replacement, or estimated WTP for battery health disclosure as distinct from the battery condition itself. These gaps motivate the present study.

2.4. Willingness to Pay for Battery Information

To be continued.

3. Survey Design, Data Collection, and Sample Profile

3.1. Nationwide Used EV Survey

The survey was designed to elicit consumer preferences and WTP for vehicle attributes and battery health information in the used BEV market. The target population consists of U.S. adults aged 18 or older who reported an intention to purchase a used car or SUV within the next two years.

The centerpiece of the survey is a BEV discrete choice experiment in which respondents choose among hypothetical used BEVs with varying attributes. The DCE is designed to capture the trade-offs consumers make when evaluating used BEVs, with particular attention to how battery health information is valued relative to conventional vehicle characteristics. Each respondent completes six repeated choice tasks. Prior to the choice tasks, respondents are shown descriptions of all attributes they will encounter, including detailed definitions for battery-specific terms (see Table 1). Following prior evidence that age and mileage are primary determinants of used vehicle value (Prieto et al., 2015), the choice scenarios hold vehicle age constant: all vehicles are assumed to have been manufactured in 2022 and to be three years old at the time of purchase. The varying attributes are vehicle mileage, purchase price, battery refurbishment history, electric range, and battery SOH.

During the DCE design phase, we used range at the time of manufacture (Year 0) and an annual degradation rate to characterize battery condition. To improve comprehension for respondents, we transformed these underlying design parameters and presented range and SOH at Year 3 (the current purchase date) and Year 8 (five years after purchase). We tested several attribute specifications and selected the one that combines current range with the proportional rate of range loss between Year 3 and Year 8 on the basis of model fit and economic interpretability. This specification outperformed alternatives that used absolute range loss or range at Year 0, suggesting that consumers attend more to the relative deterioration of battery health over their expected ownership window than to the vehicle's original specifications.

To examine how information provision shapes preferences, respondents were randomly assigned to one of two treatment conditions. The *basic information* group received descriptions of electric range and battery SOH only. The *extended information* group additionally received information about battery maintenance history and warranty coverage remaining on the vehicle. This design allows us to test whether the content of battery information affects consumers' choices and their propensity to opt out. Figure 1 shows an example choice task, in which respondents select among three used BEVs or a "None of the above" option.

Table 1
BEV choice experiment attributes and levels.

Attribute	Definition	Levels			
		Low-budget (≤$20K) Car	**High-budget** (>$20K) Car	**Low-budget** (≤$20K) SUV	**High-budget** (>$20K) SUV
Mileage	The total number of miles a vehicle has traveled since manufacture.	15,000–60,000 miles, in increments of 500 miles.			
Battery refurbishment history	The repair or replacement work that has been performed on the vehicle's main battery pack.	*(1) Original:* The battery remains in its factory-original condition with no repairs or replacements. *(2) Some battery cells replaced:* A portion of the battery cells has been replaced to address performance issues or extend battery life; the remaining cells are original. *(3) Entire battery pack replaced:* The full battery pack has been replaced with a new, refurbished, or used pack, typically after battery performance declined below a specified threshold.			
Electric range at delivery (Year 0)	The maximum distance the vehicle can travel on a full battery charge under typical driving conditions, as rated at the time of manufacture.	50–150 miles (50-mile increments)	100–250 miles (50-mile increments)	150–250 miles (50-mile increments)	200–350 miles (50-mile increments)
Battery annual degradation rate and state of health (SOH)	The average annual percentage loss in electric range and battery SOH. The remaining range and SOH at Year 3 (current) and Year 8 (five years from purchase) are derived from this rate.	1%–8%, in increments of 1%.			
Purchase price	The total cost of the vehicle in dollars, including down payment, monthly payments, taxes, and fees.	$10,000–$20,000 ($2,000 increments)	$20,000–$40,000 ($5,000 increments)	$15,000–$25,000 ($2,000 increments)	$25,000–$45,000 ($5,000 increments)

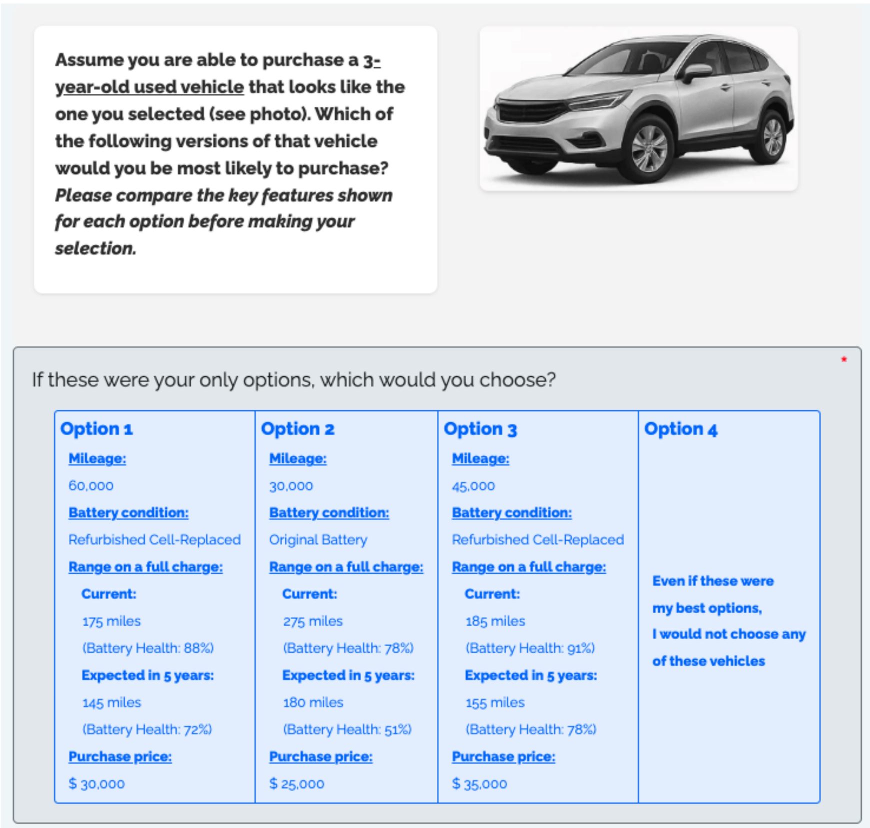

Figure 1: Example of a choice experiment.

In addition to the DCE, the survey collects information across three broader topic areas. Before the choice tasks, respondents report on their current household vehicle fleet—including fuel type, range, acquisition method, and costs—and their future vehicle purchasing plans, covering anticipated budget, payment method, and the likelihood of purchasing a new or used plug-in hybrid or BEV. After the choice tasks, the survey assesses respondents' knowledge, attitudes, and perceptions regarding EVs and battery technology. This section includes factual questions about EV charging capabilities and available federal tax incentives, as well as a battery of attitudinal statements rated on five-point Likert scales. The attitudinal items cover social norms, environmental beliefs, cost perceptions, battery concerns, and personality traits such as risk-taking and price sensitivity. Two attitudinal items about refurbished EV batteries are of particular analytical relevance: "Purchasing refurbished EV batteries will minimize negative effects on natural ecosystems" (hereafter "environmentally positive") and "Refurbished EV batteries do NOT perform and function as original EV batteries" (hereafter "functionally negative"). The final survey section collects demographic information including age, gender, race and ethnicity, household size, income, education, employment status, and housing type.

For a full description of the survey instrument, see Gore et al. (2026).[7]

3.2. Data Collection and Sample Profile

The survey was administered online through two widely-used research platforms between December 2025 and April 2026. The median completion time was approximately 14 minutes. All participants received platform-standard compensation.

The initial sample comprised 3,657 respondents from across the United States. We applied multiple data quality screens to identify and remove low-quality responses, including speeders, straightliners on the DCE tasks, respondents who failed the embedded attention check, those with internal inconsistencies across survey sections, and those who provided uninformative or incoherent text in open-ended responses. The final analytic sample consists of 2,916 respondents, yielding 17,496 choice observations across six tasks. Figure 2 shows the geographic distribution of respondents by ZIP code; point size and color intensity are scaled to respondent count, indicating broad national coverage.

[7] The Data Collection Instrument was published prior to data collection. Certain aspects of the survey design were revised following piloting. Where discrepancies exist between the instrument and this paper, the description in this paper reflects the final implemented version.

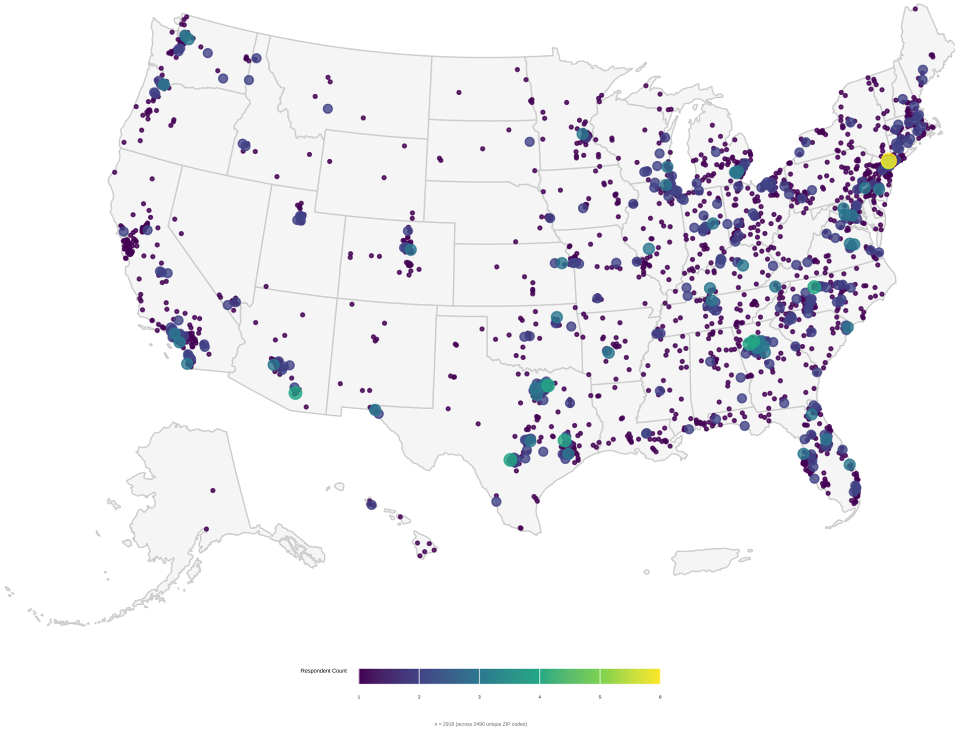

Figure 2: Geographic distribution of survey respondents across the continental United States.

4. Method

4.1. Mixed Logit Model

We begin by estimating WTP for changes in vehicle attributes across the full sample, treating the population as a single group. This provides an aggregate baseline and reveals the extent of preference heterogeneity. We estimate a Mixed Logit (MXL) model directly in *WTP space*. The MXL model relaxes the Independence of Irrelevant Alternatives (IIA) assumption of the standard Multinomial Logit (MNL) model by allowing preference parameters to vary continuously across individuals (McFadden and Train, 2000; Train, 2009). Estimating in WTP space allows us to directly specify the distribution of WTP for each non-price attribute, rather than recovering it post hoc from the ratio of attribute to price coefficients. We assume that individual WTP values follow normal distributions, capturing symmetric heterogeneity around population means. The utility parameterization follows the notation in Helveston et al. (2018) and Helveston (2023).

The WTP-space utility for respondent n choosing alternative i in choice scenario t is:

$$u_{nti} = \lambda_n \left(\boldsymbol{\omega}_n^\top \mathbf{x}_{nti} - p_{nti} \right) + \varepsilon_{nti}, \qquad \varepsilon_{nti} \sim \text{Gumbel}\left(0, \frac{\pi^2}{6}\right) \tag{1}$$

where p_{nti} is the purchase price presented in scenario t to respondent n; $\mathbf{x}_{nti}$ is a vector of non-price BEV attributes including vehicle mileage, battery refurbishment status, current electric range at the point of purchase (Year 3), and the proportional rate of range loss over the subsequent five years (to Year 8);[8] $\lambda_n > 0$ is the individual-specific scale parameter; $\boldsymbol{\omega}_n$ is a vector of individual WTP values drawn from a mixing distribution $f(\boldsymbol{\omega} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma})$ with population mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$; and ε_{nti} is an IID Type I Extreme Value error. Because λ_n and $\boldsymbol{\omega}_n$ are both unobserved, the choice probability requires integrating the logit kernel over their joint mixing distribution:

[8] We compared six attribute specifications that varied whether range was expressed at Year 0 or Year 3, and whether battery health change was expressed as absolute range loss or as a percentage degradation rate. The specification combining range at Year 3 with percentage loss rate from Year 3 to Year 8 yielded the best model fit and the most economically interpretable coefficients, suggesting consumers evaluate used BEVs primarily in terms of current performance and relative future deterioration.

$$P_{nti}(\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \int\int \frac{\exp\big(\lambda\big(\boldsymbol{\omega}^\top \mathbf{x}_{nti} - p_{nti}\big)\big)}{\sum_j \exp\big(\lambda\big(\boldsymbol{\omega}^\top \mathbf{x}_{ntj} - p_{ntj}\big)\big)} f(\boldsymbol{\omega} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma})\, g(\lambda)\, d\boldsymbol{\omega}\, d\lambda \tag{2}$$

This integral is evaluated by simulation using Halton draws ($R = 500$). Accounting for the panel structure of six repeated choice tasks per respondent, the simulated log-likelihood is:

$$\mathcal{L}(\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \sum_{n=1}^{N} \ln\left[\frac{1}{R}\sum_{r=1}^{R}\prod_{t=1}^{T} \frac{\exp\Big(\lambda_r\big(\boldsymbol{\omega}_r^\top \mathbf{x}_{nti^*_{nt}} - p_{nti^*_{nt}}\big)\Big)}{\sum_j \exp\big(\lambda_r\big(\boldsymbol{\omega}_r^\top \mathbf{x}_{ntj} - p_{ntj}\big)\big)}\right] \tag{3}$$

where $(\lambda_r, \boldsymbol{\omega}_r)$ is the r-th draw from the joint mixing distribution and i^*_{nt} is the alternative chosen by respondent n in scenario t. Because estimation is carried out directly in WTP space, the population mean WTP for each non-price attribute k is obtained directly as $\hat{\mu}_k$, without requiring a post-estimation ratio transformation. We estimate the model using the *logitr* R package (Helveston, 2023).

4.2. Latent Class Choice Model

Model Framework

The MXL results reveal substantial preference heterogeneity across respondents. To explore whether this heterogeneity reflects discrete consumer segments with systematically different preferences, we estimate a Latent Class Choice Model (LCCM). The LCCM assumes that individuals can be assigned to a finite number of unobserved groups (classes) within which preferences are homogeneous, but across which they differ (Greene and Hensher, 2003; Kamakura and Russell, 1989). This structure is well suited to our research context, where consumers may differ fundamentally in their attitudes toward battery health risk, their familiarity with EV technology, and their price sensitivity.

As illustrated in Figure 3, the LCCM consists of two simultaneously estimated components. The *class membership model* assigns each respondent probabilistically to a latent class as a function of observable individual characteristics. The *class-specific choice model* then estimates the conditional probability of each vehicle choice within each class. In addition, a set of *inactive indicators* — attitudinal and behavioral variables excluded from estimation due to potential collinearity or theoretical considerations — are used to further characterize the classes descriptively after estimation. The formal model structure follows the notation of Chen et al. (2023).

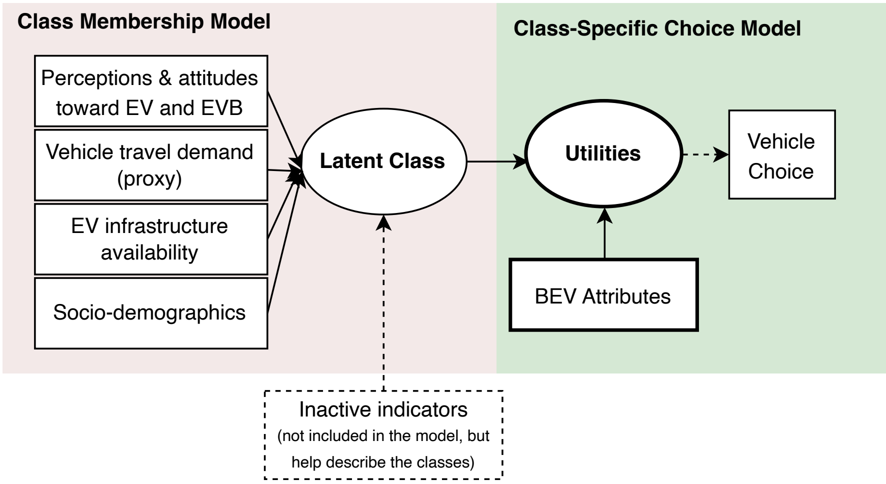

Figure 3: LCCM model framework.

Class-specific Choice Model

For respondent n belonging to latent class s ($s = 1, 2, \ldots, S$), the utility of choosing vehicle alternative i in choice scenario t ($t = 1, \ldots, 6$) is:

$$U_{nti|s} = \boldsymbol{\beta}_s' \mathbf{X}_{nti} + \varepsilon_{nti|s} \tag{4}$$

where $\mathbf{X}_{nti}$ is a vector of observed BEV attributes (vehicle mileage, battery refurbishment status, electric range at Year 3, proportional range loss to Year 8, and purchase price); $\boldsymbol{\beta}_s$ is a vector of class-specific utility parameters; and $\varepsilon_{nti|s}$ is an IID Type I Extreme Value error. Under this assumption, the conditional probability that respondent n in class s chooses alternative i in scenario t follows the MNL form:

$$P_{nt}(i \mid s) = \frac{\exp\left(\boldsymbol{\beta}_s' \mathbf{X}_{nti}\right)}{\sum_{j \in C_{nt}} \exp\left(\boldsymbol{\beta}_s' \mathbf{X}_{ntj}\right)} \tag{5}$$

where C_{nt} denotes the choice set available to respondent n at scenario t.

To capture the non-linear relationship between range and choice probability, we adopt a piecewise linear specification for the range and degradation rate attributes. Consumers likely have diminishing sensitivity to range at higher levels and heightened sensitivity around critical viability thresholds. We compared a quadratic specification, which imposes a symmetric parabolic shape and performs poorly at the tails, with a piecewise linear specification that estimates separate slopes for distinct segments. The piecewise specification provides superior model fit and more interpretable coefficients. The breakpoints are defined as follows: electric range at Year 3 is segmented into 40–130 miles, 130–200 miles, and above 200 miles; the range loss rate is segmented into below 12%, 12–24%, and above 24%. These thresholds reflect both theoretical considerations about minimum driving viability and empirical patterns in the data.

Class Membership Model

The class membership model specifies the probability that respondent n belongs to latent class s as a function of observed individual characteristics:

$$M_n(s) = \frac{\exp\left(\boldsymbol{\gamma}'_s \mathbf{Z}_n\right)}{\sum_{s'=1}^{S} \exp\left(\boldsymbol{\gamma}'_{s'} \mathbf{Z}_n\right)} \tag{6}$$

where $\mathbf{Z}_n$ is a vector of individual-level covariates organized into four groups: (1) *perceptions and attitudes toward EVs and EV batteries* (e.g., range anxiety, environmental and functional assessments of refurbished batteries); (2) *vehicle travel demand proxies* (e.g., range of the respondent's primary vehicle); (3) *EV infrastructure availability* (e.g., access to electrical outlets for charging); and (4) *socio-demographic characteristics* (e.g., household income, vehicle type preference). The parameters for the reference class S are normalized to zero for identification.

The unconditional probability that respondent n chooses alternative i in scenario t, integrating over all latent classes, is:

$$P_{nt}(y_n = i) = \sum_{s=1}^{S} P_{nt}(i \mid s) \cdot M_n(s) \tag{7}$$

Substituting Equations (5) and (6) into Equation (7) gives the full integrated choice probability:

$$P_{nt}(y_n = i) = \sum_{s=1}^{S} \left[\frac{\exp\left(\boldsymbol{\beta}'_s \mathbf{X}_{nti}\right)}{\sum_{j \in C_{nt}} \exp\left(\boldsymbol{\beta}'_s \mathbf{X}_{ntj}\right)} \cdot \frac{\exp\left(\boldsymbol{\gamma}'_s \mathbf{Z}_n\right)}{\sum_{s'=1}^{S} \exp\left(\boldsymbol{\gamma}'_{s'} \mathbf{Z}_n\right)} \right] \tag{8}$$

The log-likelihood across all N respondents and T choice scenarios is:

$$\mathcal{L}(\boldsymbol{\beta}, \boldsymbol{\gamma}) = \sum_{n=1}^{N} \sum_{t=1}^{T} \ln P_{nt}(y_n = i_n^*) \tag{9}$$

where i_n^* denotes the alternative chosen by respondent n in scenario t. The model parameters $\boldsymbol{\beta}_s$ and $\boldsymbol{\gamma}_s$ are estimated jointly via maximum likelihood.

Willingness to Pay

A key output of the LCCM is class-specific WTP for each BEV attribute. Under the assumption that the marginal utility of price is negative and constant across alternatives, the WTP for attribute k in class s is:

$$\mathrm{WTP}_{s,k} = -\frac{\hat{\beta}_{s,k}}{\hat{\beta}_{s,\mathrm{price}}} \tag{10}$$

This represents the maximum dollar amount a representative respondent in class s would pay, or accept as compensation, for a one-unit change in attribute k, holding all other attributes constant.

5. Results and Discussion

5.1. Results of Mixed Logit Model

Table 2 presents the MXL estimates. All mean WTP parameters are significant at the 0.1% level (***). On average, respondents value an additional 100 miles of electric range at Year 3 by \$11.2k, penalize every 10,000 miles of odometer reading by \$2.8k, and discount each additional percentage point of range loss rate by \$1.0k. Both refurbishment types generate meaningful disutility: pack replacement reduces WTP by \$4.0k and cell replacement by \$4.5k. The no-choice opt-out option carries a large negative coefficient (\$51.9k), reflecting a strong baseline preference to purchase a vehicle rather than opt out.

The standard deviation estimates confirm substantial preference heterogeneity across respondents. Range at Year 3 exhibits the widest dispersion in absolute terms (SD = \$14.0k relative to a mean of \$11.1k), with the distribution of simulated WTP draws spanning from \$1.7k at the first quartile to \$20.5k at the third quartile. This indicates that while most respondents positively value range, a non-trivial share places little weight on it. Heterogeneity in the refurbishment parameters is also pronounced: the standard deviations for pack replacement and cell replacement are both larger in magnitude than their respective means, and the upper quartiles of the simulated distributions are positive (\$1.1k and \$0.1k, respectively). This implies that roughly one quarter of consumers view battery refurbishment neutrally or even as a positive quality signal, potentially because a replaced battery is perceived as newer than the original. Mileage dispersion is also substantial, with approximately one quarter of the simulated distribution crossing zero. In contrast, range loss rate is the most uniformly penalized attribute: its standard deviation is small relative to the mean, and the distribution remains negative throughout nearly the entire sample. Together, these results confirm that consumer valuations of battery-related attributes are heterogeneous in both sign and magnitude, motivating the segment-level analysis that follows.

Table 2
Mixed Logit Model Estimates in WTP Space (price unit: \$10,000).

Parameter	Estimate	Std. Error	*p*-value				
Mean parameters					*Summary of 10k Draws*		
					1st Qu.	Median	3rd Qu.
λ (scale)	0.980	0.025	<0.001	***			
Mileage	-0.280	0.015	<0.001	***	-0.552	-0.280	-0.008
Range (Year 3)	1.114	0.046	<0.001	***	0.172	1.113	2.054
Range loss Rate (Year 3 to Year 8, in percentage)	-0.100	0.003	<0.001	***	-0.153	-0.100	-0.047
Battery Refurbishment: Pack Replace	-0.401	0.038	<0.001	***	-0.911	-0.402	0.106
Battery Refurbishment: Cell Replace	-0.448	0.038	<0.001	***	-0.904	-0.447	0.011
No choice (opt-out)	-5.187	0.114	<0.001	***			
Standard deviation							
SD: Mileage	-0.404	0.017	<0.001	***			
SD: Range (Year 3)	1.396	0.055	<0.001	***			
SD: Range loss Rate (Year 3 to Year 8, in percentage)	-0.079	0.003	<0.001	***			
SD: Battery Refurbishment: Pack Replace	0.754	0.061	<0.001	***			
SD: Battery Refurbishment: Cell Replace	-0.678	0.063	<0.001	***			
Observations							$2{,}916 \times 6 = 17{,}496$
Parameters							12
Log-Likelihood							-17,647.52
Null Log-Likelihood							-24,254.61
AIC							35,319.03
BIC							35,412.27
McFadden R^2							0.2724
Adj. McFadden R^2							0.2719

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1.

5.2. Results of Latent Class Choice Model

5.2.1. Model Selection

To determine the optimal number of latent classes, we estimated models with one through eight classes (Table 3). Model fit was evaluated using log-likelihood (LL), the ρ^2 measure relative to the null model, Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), adjusted ρ^2, and entropy, where smaller AIC and BIC values and larger adjusted ρ^2 and entropy indicate better fit and class separation (Nylund et al., 2007). We selected a six-class solution on the basis of fit statistics and interpretability. In the seven- and eight-class solutions, one class consistently accounted for fewer than 6% of the sample, reducing robustness for segment-level analysis.

Table 3
Latent Class Model Fit Index Comparison.

No. of Classes	Npar	LL	L^2	AIC	BIC	Adj.ρ^2	Entropy	Class proportions
1	11	-20,474	0	40,970	41,056	0.148	—	100.0%
2	33	-17,917	5,114	35,901	36,157	0.254	0.911	78.8% / 21.2%
3	55	-17,411	6,126	34,931	35,358	0.274	0.759	56.0% / 23.4% / 20.6%
4	77	-17,071	6,806	34,296	34,895	0.287	0.758	47.7% / 21.9% / 21.6% / 8.8%
5	99	-16,771	7,406	33,740	34,510	0.299	0.723	25.8% / 22.9% / 21.5% / 21.1% / 8.7%
6	121	-16,538	7,872	33,318	34,258	0.308	0.730	22.7% / 20.8% / 18.5% / 16.6% / 12.8% / 8.7%
7	143	-16,413	8,122	33,113	34,224	0.312	0.736	22.2% / 18.3% / 16.5% / 15.9% / 12.6% / 8.8% / 5.6%
8	165	-16,308	8,332	32,947	34,229	0.316	0.728	20.4% / 16.1% / 16.0% / 15.1% / 12.8% / 7.9% / 5.8% / 5.8%

[1] $L^2 = -2(\mathrm{LL}_{1c} - \mathrm{LL}_{\mathrm{model}})$.
[2] Adj.ρ^2 = adjusted rho-squared vs. observed shares (constants model).
[3] Entropy $= 1 - H/H_{\max}$, where $H = -\sum \pi_{ik} \log(\pi_{ik})$. Values closer to 1 indicate cleaner class separation.

5.2.2. WTP and Class Profiles

Estimated model coefficients are presented in Table 4, with **Class 1** serving as the reference class. WTP estimates and class profile characteristics are reported in Table 5, with additional metadata on opt-out behavior and survey engagement in Table 6. Figure 4 provides a visual summary of WTP patterns and segment profiles across classes. Class membership ranges from 8.7% to 22.7% of the sample. We describe each class below, organized to highlight contrasts in battery quality sensitivity, price responsiveness,market readiness, and segment-specific market strategies.

Class 1: Opt-Out Dominant, BEV-Skeptical Consumers (*n*=252, 8.7%).
Class 1 is defined primarily by systematic opt-out behavior: 75.4% of members selected the no-choice option in all six tasks, and an additional 19.4% did so in five of six. As a result, most WTP estimates for vehicle attributes are statistically indistinguishable from zero, with the exception of a moderate disutility for range loss in the 5–12% segment ($1.6k* per percentage point) and significant penalties for both pack replacement ($6.4k*) and cell replacement ($7.3k*). These patterns suggest that respondents who systematically avoid the BEV market retain particular concerns about battery quality.

Class 1 represents consumers who are not yet viable participants in the used BEV market. The majority (70.4%) strongly disagree that they are likely to purchase a used BEV. They are the oldest class on average (47.1 years old), the least risk-tolerant (24.0%), and with the least environmental awareness. Their household fleets are almost exclusively internal combustion engine vehicles (ICEVs), with 89.3% of households reporting no EVs and 93.3% driving an ICEV as their primary vehicle. A majority (68.2%) intend their next vehicle to be an SUV. EV-related knowledge is low (68.5% correctly answer the EV knowledge question; only 10.6% are aware of federal EV subsidies), while range anxiety is the highest of any class (87.2%). Attitudes toward refurbished batteries are notably negative: only 30.5% agree that they are environmentally beneficial, and just 9.1% disagree that they are functionally inferior. Infrastructure readiness is also limited: only 32.8% have home electrical outlet access, and only 24.7% report a neighbor who owns or leases a BEV or PHEV. Overall, this class is largely outside the addressable market for used BEVs in the near term.

Class 3: Range-Focused, EV-Engaged Consumers (*n*=485, 16.6%).
Class 3 stands in sharp contrast to Class 1 and is described next to highlight this polarity. It is distinguished by an exceptionally high WTP for range: $38.7k*** per 100 miles for vehicles with 40–130 miles of range, declining to $20.8k*** (130–200 miles) and $25.5k*** (200+ miles). This convex pattern suggests that consumers in this class place a higher premium either on lower-range vehicles achiving a minimum viable range threshold (which may reflect their interest in a secondary vehicle that are cost-effective and intended primarily for local trips), or on vehicles that with very high ranges (which offer flexibility and reliability for long distance travel). Consistent with this interpretation, members of this class report the highest average range for their current primary vehicle (321.3 miles), suggesting relatively demanding travel needs and greater sensitivity to vehicle range limitations. Their range loss aversion is moderate and uniformly significant across all degradation segments ($0.9k***, $0.7k***, $0.5k***). Notably, WTP for refurbishment is negligible and statistically insignificant, suggesting that this class focuses more on current and future performance rather than battery refuishment history.

Class 3 represents the most EV-engaged segment in the sample. They hold the most favorable attitudes toward refurbished batteries (65.5% agree they are environmentally beneficial; 35.5% disagree that they are functionally inferior). Their households show the lowest rate of ICEV-only composition (74.1%), and they are the most likely class to prefer a car over an SUV (57.2%). They have the highest electrical outlet access (57.5%), along with the strongest EV knowledge (81.1%) and subsidy awareness (31.9%). They have the highest household income ($97.1k), report the highest employment rate (71.5%), are the most risk-tolerant of all classes (39.0% agree with a risk-taking characterization).

For this segment, auto dealers can filter low-range inventory aggressively or provide clear range upgrade pathways. Their minimal disutility for refurbishment history, combined with high income, high EV knowledge, and positive attitudes toward refurbished batteries, suggests they are open to technically sound refurbishments provided the resulting vehicle meets their range threshold. Performance-oriented marketing that leads with current range and degradation trajectory, rather than battery service history, is most likely to resonate.

Class 2: Battery Health Attentives (*n*=661, 22.7%).

Class 2 is the largest class, which is uniquely defined by their battery health sensitivity. Range loss aversion is the highest across classes and uniform at approximately $2.3k*** per percentage point across all three degradation segments. Both refurbishment types also generate meaningful disutility ($2.7k* for pack replacement; $3.8k** for cell replacement). Their range WTP is consistent across the three piecewise segments ($4.6k, $4.6k, and $5.3k**), indicating a steady marginal valuation of additional range without a strong threshold effect though only the highest tier reaches statistical significance.

Members of Class 2 have an average household income of $91.3k and are relatively young (36.5 years). They demonstrate above-average EV knowledge (76.2%), moderate electrical outlet access (46.0%), and a high rate of EV-owning neighbors (43.5%). This class represents well-informed consumers are not necessarily purchasing range; rather, they are purchasing battery reliability at point of sale and protection against future performance loss.

Their valuation structure is well suited to transparent, attribute-based pricing. Financing products that tie interest rates or warranty terms to battery SOH readings, or that offer extended coverage contingent on passing a minimum degradation threshold, could be particularly attractive to this segment. Their strong aversion to range loss ($2.3k per percentage point across all degradation levels) also suggests that short-term ownership scenarios with rapid degradation are significantly discounted, pointing to the value of long-term degradation forecasts in the listing description.

Class 5: Multi-Attribute Maximizers (*n*=606, 20.8%).

Class 5 exhibits statistically significant evaluations across all vehicle attributes. Like Class 3, this class places a high premium on crossing the minimum viable range threshold, with still-substantial but diminishing WTP at higher range levels ($11.8k***, $9.5k***, $9.5k***). Range loss aversion is uniformly significant at $0.8k*** per percentage point across the lower two degradation segments, tapering at the highest degradation level ($0.4k*). Refurbishment disutility is among the highest of the attribute-sensitive classes ($5.8k*** for pack replacement; $6.1k*** for cell replacement).

What further distinguishes Class 5 is its opt-out pattern: only 5.2% of members never opt out, while the plurality opt out two (29.0%) or three (27.2%) times out of six tasks. Class 5 members exhibit high range anxiety (83.4%) and are particularly averse to battery refurbishment (only 18.8% disagree that refurbished batteries are functionally inferior). They also report average household incomes ($86.9k) and lower risk tolerance. These characteristics lead them to make cautious, holistic evaluations across all vehicle attributes simultaneously. The uniform magnitude and significance of their attribute benefits and penalties mean that no single attribute can easily offset large deficits elsewhere. Combined with their high opt-out frequency, this suggests that when a choice set presents no option that reach minimal quality threshold across all dimensions at once, respondents in this class choose to opt out rather than settle for the least-bad alternative.

This segment is not amenable to price discounting as a close tactic: if a vehicle does not clear the quality bar on all relevant dimensions, a lower price is unlikely to convert. Product curation and clear multi-dimensional disclosure — range, degradation rate, and refurbishment status presented together — are more effective approaches for this group.

Class 6: Budget-constrained, Low-WTP Consumers (*n*=373, 12.8%)

Class 6 has the largest price coefficient in absolute value (−4.017***), more than twice the magnitude of the next-most price-sensitive class, suggesting they are budget-constrained buyers Range WTPs are modest but statistically significant at lower range segments ($2.8k** for 40–130 miles; $1.8k* for 130–200 miles), and range loss aversion is marginal ($0.2k** and $0.2k*** in the upper degradation segments). WTP for refurbishment is negligible and statistically insignificant, suggesting that battery service history is largely immaterial in this group's purchasing calculus.

Class 6 members have the lowest average household income ($70.0k) and the lowest next-vehicle budget ($19.7k). They have the lowest homeownership rate (43.0%) and the highest share of renters (48.8%), the fewest household vehicles (1.9 on average), and the highest share of ICEV-only households (87.1%). Despite their strong price sensitivity, 91.2% of Class 6 members never select the no-choice option, indicating their interest in used vehicles, but thieir WTP is dominated by price rather than battery quality.

For this class, affordable pricing strategies and accessible financing, including buy-here-pay-here programs, certified pre-owned tiers with minimum quality floors, or OEM-backed affordability programs, are most relevant for serving this segment. Expanding used BEV access to lower-income, renter households hinges largely on whether vehicles in the $15k–$20k range can be made commercially viable.

Class 4: Price-Insensitive, Mixed-Motivation Evaluators (*n*=539, 18.5%).
The most prominent feature of this class is its exceptionally low price coefficient in absolute value (−0.144**), indicating substantially lower price sensitivity than all other classes; as a result, all calutated WTP values are ratio-inflated. We suspect this class emcompasses several several subsets of respondents with different motivations who nonetheless produce similar WTP patterns.

Class 4 exhibits the largest disutility for battery refurbishment of any class: $26.6k** for pack replacement and $32.6k** for cell replacement. One contributing factor may be that that Class 4 has the highest share of respondents who received the extended information treatment (52.7%), which included more details about battery maintenance costs and warranty coverage. This information may have heightened respondents' attention to the refurbishment attribute and amplified their concern about non-original battery conditions.

At the same time, Class 4 exhibits characteristics commonly associated with experienced EV consumers. Class 4 has the highest rates of current BEV ownership in theirhousehold (13.5%), PHEV or HEV ownership (25.3%), and EV-owning neighbors (50.9%). They report the lowest range anxiety (67.0%) and the highest risk-taking propensity (51.9%), and they have set the highest budget for their next vehicle ($29.3k). These patterns suggest that at least some respondents in this class are highly committed to EV technology and may view many of the battery-health attributes included in the experiment as less relevant to their purchase decisions.

A third explanation relates to respondent engagement. Despite detailed attribute descriptions and explanatory materials, some respondents may not have fully understood the battery-health information presented in the choice tasks. Mileage appears to be the only attribute that generated a large and statistically significant valuation ($10.0k** per 10,000 miles), whereas most other attributes produced imprecise estimates. This pattern is consistent with the possibility that some respondents focused on the most familiar vehicle characteristic while devoting limited attention to the remaining attributes. Such behavior can generate unstable parameter estimates and exaggerated WTP values that do not necessarily reflect underlying preferences. Supporting this interpretation, this class has a short completion time (on average 81 seconds across six tasks). This raises the possibility that some members of this class did not carefully evalute all choice attributes, potentially contributing to the imprecise estimates in majority of the attributes.

Taken together, Class 4 may represent a mixture of refurbishment-averse consumers, experienced EV buyers, and attribute-disengaged respondents. Although these groups likely differ in their underlying motivations, they generate a common empirical pattern characterized by very large negative valuations for battery refurbishment alongside limited sensitivity to most other battery-health attributes. As a result, the class is best interpreted cautiously, with consideration given to both information-treatment effects and potential heterogeneity in respondent engagement.

Figure 4: WTP and class profile summary.

Table 4
LCCM model results

	Class 1	Class 2	Class 3	Class 4	Class 5	Class 6
Vehicle Attributes						
No-Choice Option (opt-out)	-2.577 (1.344).	-9.819 (0.813)***	-1.968 (0.75)**	-3.491 (0.457)***	-2.513 (0.387)***	-13.847 (1.661)***
Mileage (10,000 miles)	-0.472 (0.248).	-0.3 (0.041)***	-0.31 (0.058)***	-0.143 (0.03)***	-0.22 (0.031)***	-0.229 (0.055)***
BEV Electric Range (Year 3): <130 miles	0.114 (1.326)	0.468 (0.281).	5.742 (0.417)***	0.177 (0.225)	1.426 (0.365)***	1.128 (0.399)**
BEV Electric Range (Year 3): 130-200 miles	1.884 (1.383)	0.472 (0.245).	3.089 (0.471)***	-0.145 (0.176)	1.14 (0.22)***	0.707 (0.357)*
BEV Electric Range (Year 3): 200+ miles	1.347 (0.61)*	0.545 (0.189)**	3.78 (0.733)***	0.198 (0.116).	1.145 (0.166)***	0.794 (0.507)
Range Loss Rate: <12%	-0.237 (0.072)***	-0.236 (0.029)***	-0.129 (0.024)***	0.004 (0.015)	-0.093 (0.017)***	-0.084 (0.03)**
Range Loss Rate: 12%-24%	-0.044 (0.063)	-0.239 (0.019)***	-0.102 (0.016)***	-0.018 (0.01).	-0.091 (0.013)***	-0.069 (0.02)***
Range Loss Rate: 24%+	0.035 (0.087)	-0.22 (0.03)***	-0.069 (0.019)***	-0.02 (0.01)*	-0.047 (0.019)*	-0.067 (0.02)***
Battery Refurbishment: Pack Replace	-0.932 (0.431)*	-0.281 (0.126)*	-0.08 (0.159)	-0.384 (0.105)***	-0.695 (0.106)***	0.041 (0.166)
Battery Refurbishment: Cell Replace	-1.074 (0.417)*	-0.385 (0.127)**	-0.099 (0.145)	-0.47 (0.102)***	-0.732 (0.11)***	0.109 (0.165)
Purchase Price (10,000 USD)	-1.467 (0.652)*	-1.025 (0.118)***	-1.484 (0.287)***	-0.144 (0.054)**	-1.206 (0.1)***	-4.017 (0.363)***
Active Indicators						
ASC	0	0.867 (0.5).	-0.272 (0.552)	2.641 (0.493)***	1.379 (0.539)*	1.28 (0.577)*
Perceived EV Range Anxiety: Agree	0	-0.021 (0.257)	-0.131 (0.272)	-0.734 (0.251)**	-0.118 (0.269)	-0.513 (0.283).
Risk Taking Propensity: Agree	0	0.692 (0.2)***	0.387 (0.214).	0.893 (0.204)***	0.042 (0.207)	0.001 (0.242)
Household Income (10,000 USD)	0	0.011 (0.016)	0.021 (0.018)	0.003 (0.017)	0 (0.018)	-0.048 (0.022)*
Electrical Outlet Access	0	0.391 (0.191)*	0.812 (0.205)***	0.525 (0.197)**	0.305 (0.195)	0.423 (0.223).
EV Battery Knowledge: Yes	0	0.146 (0.194)	0.307 (0.225)	-0.183 (0.202)	0.284 (0.197)	0.009 (0.223)
EV Battery Environmentally Positive: Agree	0	1.209 (0.189)***	1.295 (0.211)***	0.808 (0.196)***	0.829 (0.191)***	1.186 (0.217)***
EV Battery Functionally Negative: Disagree	0	0.987 (0.278)***	1.547 (0.284)***	0.822 (0.288)**	0.745 (0.29)*	1.169 (0.303)***
Current Primary Household Vehicle Fuel Type: ICEV	0	-0.629 (0.268)*	-0.592 (0.279)*	-1.227 (0.261)***	-0.596 (0.268)*	-0.015 (0.332)
Current Primary Vehicle Typical Range (miles)	0	-0.173 (0.091).	0.076 (0.098)	-0.325 (0.094)***	-0.113 (0.096)	-0.186 (0.101).
SUV/Crossover as the Next Vehicle	0	-0.203 (0.191)	-0.984 (0.273)***	-0.52 (0.198)**	-0.531 (0.191)**	-0.736 (0.222)***

Est. (SE)[Sig.]
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1.

Table 5
WTP and Characteristics by Class

	Class 1 (n=252, 8.7%)	Class 2 (n=661, 22.7%)	Class 3 (n=485, 16.6%)	Class 4 (n=539, 18.5%)	Class 5 (n=606, 20.8%)	Class 6 (n=373, 12.8%)
WTP (*1000 USD) for Vehicle Attributes						
No-Choice Option (opt-out)	-$17.6.	-$95.8***	-$13.3**	-$242.4**	-$20.8***	-$34.5***
Mileage (10,000 miles)	-$3.2.	-$2.9***	-$2.1***	-$10.0**	-$1.8***	-$0.6***
BEV Range: 40-130 mi segment (per 100 miles)	$0.8	$4.6.	$38.7***	$12.3	$11.8***	$2.8**
BEV Range: 130-200 mi segment (per 100 miles)	$12.8	$4.6.	$20.8***	-$10.1	$9.5***	$1.8*
BEV Range: 200+ mi segment (per 100 miles)	$9.2*	$5.3**	$25.5***	$13.8.	$9.5***	$2.0
Range Loss Rate: 5-12% (per %)	-$1.6*	-$2.3***	-$0.9***	$0.3	-$0.8***	-$0.2**
Range Loss Rate: 12-24% (per %)	-$0.3	-$2.3***	-$0.7***	-$1.3.	-$0.8***	-$0.2***

Table 5
WTP and Characteristics by Class *(continued)*

	Class 1 (n=252, 8.7%)	Class 2 (n=661, 22.7%)	Class 3 (n=485, 16.6%)	Class 4 (n=539, 18.5%)	Class 5 (n=606, 20.8%)	Class 6 (n=373, 12.8%)
Range Loss Rate: 24%+ (per %)	$0.2	-$2.1***	-$0.5***	-$1.4*	-$0.4*	-$0.2***
Battery Refurbishment: Pack Replace	-$6.4*	-$2.7*	-$0.5	-$26.6**	-$5.8***	$0.1
Battery Refurbishment: Cell Replace	-$7.3*	-$3.8**	-$0.7	-$32.6**	-$6.1***	$0.3
Active Indicators						
Risk-taking Propensity: Agree: no	76.0%	56.0%	61.0%	48.1%	71.3%	72.4%
Risk-taking Propensity: Agree: yes	24.0%	44.0%	39.0%	51.9%	28.7%	27.6%
Perceived EV Range Anxiety: Agree: no	12.8%	17.0%	19.4%	33.0%	16.6%	22.0%
Perceived EV Range Anxiety: Agree: yes	87.2%	83.0%	80.6%	67.0%	83.4%	78.0%
EV Battery Environmentally Positive: Agree: no	69.5%	37.4%	34.5%	46.4%	47.7%	39.6%
EV Battery Environmentally Positive: Agree: yes	30.5%	62.6%	65.5%	53.6%	52.3%	60.4%
EV Battery Functionally Negative: Disagree: no	90.9%	76.9%	64.5%	78.5%	81.2%	72.1%
EV Battery Functionally Negative: Disagree: yes	9.1%	23.1%	35.5%	21.5%	18.8%	27.9%
Household Income (1000 USD)	85.5	91.3	97.1	89.3	86.9	70.0
EV Knowledge: no	31.5%	23.8%	18.9%	28.4%	23.5%	30.7%
EV Knowledge: yes	68.5%	76.2%	81.1%	71.6%	76.5%	69.3%
Electrical Outlet Access: no	63.0%	47.8%	37.2%	42.3%	51.8%	52.3%
Electrical Outlet Access: not_sure	4.1%	6.3%	5.3%	5.8%	6.4%	7.8%
Electrical Outlet Access: yes	32.8%	46.0%	57.5%	51.9%	41.8%	39.8%
Primary Vehicle Typical Range (miles)	306.8	292.2	321.3	267.6	297.4	282.2
Household Vehicle Fuel Composition: has_bev	1.7%	7.5%	9.2%	13.5%	7.5%	3.4%
Household Vehicle Fuel Composition: has_phev_hev	7.6%	16.1%	16.6%	25.3%	13.3%	9.5%
Household Vehicle Fuel Composition: icev_only	89.3%	76.2%	74.1%	61.2%	79.0%	87.1%
Household Vehicle Fuel Composition: other	1.4%	0.1%	0.0%	0.0%	0.2%	0.0%
Next Vehicle Type: SUV: no	31.8%	38.9%	57.2%	48.3%	46.3%	53.3%
Next Vehicle Type: SUV: yes	68.2%	61.1%	42.8%	51.7%	53.7%	46.7%
Inactive Indicators						
EV Subsidy Knowledge: no	89.4%	73.6%	68.1%	70.9%	78.0%	79.1%
EV Subsidy Knowledge: yes	10.6%	26.4%	31.9%	29.1%	22.0%	20.9%
Neighbor Owns/Leases a BEV/PHEV: no	46.9%	29.6%	26.0%	28.5%	31.8%	33.9%
Neighbor Owns/Leases a BEV/PHEV: not_sure	28.3%	26.9%	26.8%	20.6%	30.3%	31.2%
Neighbor Owns/Leases a BEV/PHEV: yes	24.7%	43.5%	47.2%	50.9%	37.9%	34.9%
Age	47.1	36.5	37.5	37.4	40.9	37.7
Gender: female	67.3%	65.1%	52.6%	57.9%	63.1%	60.1%
Gender: male	31.3%	32.7%	44.9%	41.3%	34.4%	36.0%
Gender: other	1.4%	1.9%	2.5%	0.8%	2.5%	3.2%
Gender: prefer_not_answer	0.0%	0.3%	0.0%	0.1%	0.0%	0.7%
Ethnicity: hispanic	5.3%	11.6%	9.8%	15.5%	11.5%	11.6%
Ethnicity: non-hispanic	94.7%	88.4%	90.2%	84.5%	88.5%	88.4%
Race: african_american_only	7.1%	17.9%	14.4%	22.3%	11.4%	11.7%
Race: other	10.4%	15.6%	14.5%	18.5%	18.1%	15.9%
Race: white_only	82.5%	66.5%	71.2%	59.2%	70.5%	72.4%
Education Level: bachelor	31.9%	36.8%	38.4%	32.9%	36.1%	31.7%
Education Level: graduate	17.0%	20.2%	20.4%	20.4%	19.6%	17.9%
Education Level: high_school	18.6%	12.6%	11.9%	16.8%	11.4%	16.0%
Education Level: prefer_not_answer	0.0%	0.2%	0.0%	0.1%	0.5%	0.5%
Education Level: some_college	32.6%	30.2%	29.3%	29.8%	32.4%	33.9%
Student Status: non-student	92.4%	84.1%	84.6%	84.0%	86.0%	80.6%
Student Status: prefer_not_answer	0.7%	1.1%	0.8%	0.9%	1.8%	1.9%
Student Status: student	6.9%	14.8%	14.6%	15.1%	12.1%	17.5%
Employment Status: full_time	38.2%	47.2%	48.4%	46.9%	43.1%	38.6%
Employment Status: not_employed	36.5%	26.0%	27.7%	29.4%	33.1%	33.5%
Employment Status: part_time	24.6%	25.7%	23.1%	22.8%	21.9%	26.1%
Employment Status: prefer_not_answer	0.7%	1.1%	0.8%	0.9%	1.8%	1.9%
Household Size	2.8	3.1	2.9	3.3	2.9	2.8
Household Tenure: other	6.1%	6.1%	5.8%	4.7%	4.9%	7.2%
Household Tenure: own	58.3%	49.6%	54.4%	49.5%	52.5%	43.0%

Table 5
WTP and Characteristics by Class *(continued)*

	Class 1 (n=252, 8.7%)	Class 2 (n=661, 22.7%)	Class 3 (n=485, 16.6%)	Class 4 (n=539, 18.5%)	Class 5 (n=606, 20.8%)	Class 6 (n=373, 12.8%)
Household Tenure: prefer_not_answer	1.6%	1.2%	1.8%	1.8%	1.6%	1.0%
Household Tenure: rent	34.1%	43.0%	38.1%	43.9%	41.0%	48.8%
Household Type: apart	17.2%	24.1%	21.6%	26.8%	21.9%	27.6%
Household Type: other	4.7%	3.2%	3.1%	3.2%	4.1%	3.9%
Household Type: sf_attached	9.2%	11.9%	9.2%	12.4%	10.6%	10.3%
Household Type: sf_detached	68.9%	60.9%	66.0%	57.6%	63.5%	58.2%
Household Vehicle Count	2.0	2.1	2.1	2.1	2.0	1.9
Primary Vehicle Fuel Type: bev	0.9%	3.1%	4.7%	6.1%	4.3%	1.4%
Primary Vehicle Fuel Type: icev	93.3%	85.3%	84.9%	72.0%	87.4%	91.6%
Primary Vehicle Fuel Type: phev_hev	5.8%	11.6%	10.4%	21.9%	8.3%	6.9%
Next Vehicle Budget (1000 USD)	24.3	27.8	26.9	29.3	25.2	19.7
Likelihood of buying new BEV: neutral	5.2%	10.7%	11.3%	14.8%	10.0%	10.0%
Likelihood of buying new BEV: somewhat_agree	5.3%	16.8%	16.7%	22.3%	13.1%	12.5%
Likelihood of buying new BEV: somewhat_disagree	10.6%	20.4%	22.2%	16.9%	16.4%	19.1%
Likelihood of buying new BEV: strongly_agree	1.7%	7.4%	5.8%	11.1%	4.6%	3.3%
Likelihood of buying new BEV: strongly_disagree	77.2%	44.6%	44.0%	34.9%	55.9%	55.2%
Likelihood of buying used BEV: neutral	5.5%	12.6%	13.2%	16.7%	12.3%	13.6%
Likelihood of buying used BEV: somewhat_agree	6.3%	26.2%	26.2%	26.7%	18.9%	23.4%
Likelihood of buying used BEV: somewhat_disagree	14.7%	20.5%	24.1%	18.0%	19.7%	22.0%
Likelihood of buying used BEV: strongly_agree	3.2%	10.5%	10.1%	11.1%	8.0%	7.5%
Likelihood of buying used BEV: strongly_disagree	70.4%	30.2%	26.4%	27.6%	41.2%	33.5%
Climate Concern: no	78.3%	56.5%	57.6%	62.4%	59.8%	58.4%
Climate Concern: yes	21.7%	43.5%	42.4%	37.6%	40.2%	41.6%

Table 6
Metadata Summary by Class

	Class 1 (n=252, 8.7%)	Class 2 (n=661, 22.7%)	Class 3 (n=485, 16.6%)	Class 4 (n=539, 18.5%)	Class 5 (n=606, 20.8%)	Class 6 (n=373, 12.8%)
Metadata: Information Treatment						
Battery Information Treatment: no	51.6%	49.3%	50.7%	47.3%	48.3%	50.3%
Battery Information Treatment: yes	48.4%	50.7%	49.3%	52.7%	51.7%	49.7%
Metadata: Opt-out Count						
Opt-out: 0 times	0.0%	86.4%	85.7%	85.0%	5.2%	91.2%
Opt-out: 1 time	0.0%	12.0%	10.7%	12.2%	18.6%	7.4%
Opt-out: 2 times	0.0%	1.6%	2.9%	2.7%	29.0%	1.1%
Opt-out: 3 times	0.4%	0.1%	0.6%	0.1%	27.2%	0.3%
Opt-out: 4 times	4.8%	0.0%	0.0%	0.0%	15.3%	0.0%
Opt-out: 5 times	19.4%	0.0%	0.0%	0.0%	4.1%	0.0%
Opt-out: 6 times	75.4%	0.0%	0.0%	0.0%	0.6%	0.0%
Metadata: Survey Duration						
Battery DCE Section: Q1	22.0	24.0	26.0	19.0	26.0	24.0
Battery DCE Section: Q2	9.0	15.0	16.0	12.0	16.0	14.0
Battery DCE Section: Q3	7.0	13.0	15.0	11.0	13.0	12.0
Battery DCE Section: Q4	7.0	12.0	14.0	10.0	12.0	12.0
Battery DCE Section: Q5	6.0	12.0	13.0	9.0	12.0	11.0
Battery DCE Section: Q6	5.0	11.0	12.0	9.0	11.0	11.0
Battery DCE Section: Total	63.0	97.0	105.0	81.0	99.0	93.0
Full Survey	884.0	905.0	897.0	880.0	876.0	815.0

5.3. Results of Opt-Out Analysis

Among respondents who selected the no-choice option in all six choice tasks (n=193; 96 in the basic information group, 97 in the extended information group), we collected open-ended self-reports explaining why they did not choose any of the presented used BEV options. We coded these responses into thematic categories and present their distribution by information treatment in Figure 5. The basic information condition includes only BEV driving range and battery health information, while the extended information condition additionally describes battery maintenance costs, warranty coverage, and expected degradation trajectories. The thematic categories are not mutually exclusive; respondents could cite multiple reasons for opting out, and the percentages reflect the share of opt-out respondents who mentioned each theme.

The most prevalent reason across both groups is general disinterest in BEVs, though it is notably more common among extended information respondents (49%) than basic information respondents (40%)—the largest single gap between treatment groups. Conventional gas vehicle enthusiasm is identical across conditions (14% each), and EV distrust is similarly comparable (12% vs. 10%).

Among specific practical barriers, several are more prevalent in the basic information group. Price and cost concerns are cited by 24% of basic information respondents compared to 14% of extended information respondents, a gap of 10 percentage points. Range anxiety (25% vs. 18%) and used vehicle distrust (10% vs. 4%) are also notably higher in the basic information group. Charging infrastructure concerns follow a similar but more modest pattern (29% vs. 24%), as do battery concerns (23% vs. 18%). General EV skepticism runs in the opposite direction, being somewhat more prevalent among extended information respondents (16% vs. 11%). Environmental doubt is rare in both groups (5% Basic, 2% Extended).

These patterns are consistent with a salience-shifting interpretation: receiving detailed battery health and warranty information appears to consolidate opt-out reasoning toward a general rejection of BEVs, while respondents with only basic information tend to articulate their reluctance through specific practical barriers—price, range, charging, and battery condition. The extended treatment does not reduce the total rate of opt-out behavior, but it changes the channel through which reluctance is expressed. This suggests that simply providing more technical information is insufficient to convert disengaged consumers into active buyers; it may instead sharpen awareness of fundamental incompatibility between the product category and the respondent's preferences or circumstances.

This finding cautions against a strategy of disclosure-as-sufficient. For market communication to convert awareness into purchase confidence, information must be paired with credible assurance. Certification programs that attach independent SOH scores to vehicle listings, warranty products that guarantee a minimum performance threshold through the ownership window, and resale guarantees that hold dealers accountable for disclosed condition all represent mechanisms that transform disclosed information into a purchasable guarantee. The financial infrastructure around battery health—secondary warranty products, SOH-adjusted financing rates, and certified resale programs—is therefore at least as commercially important as the information itself.

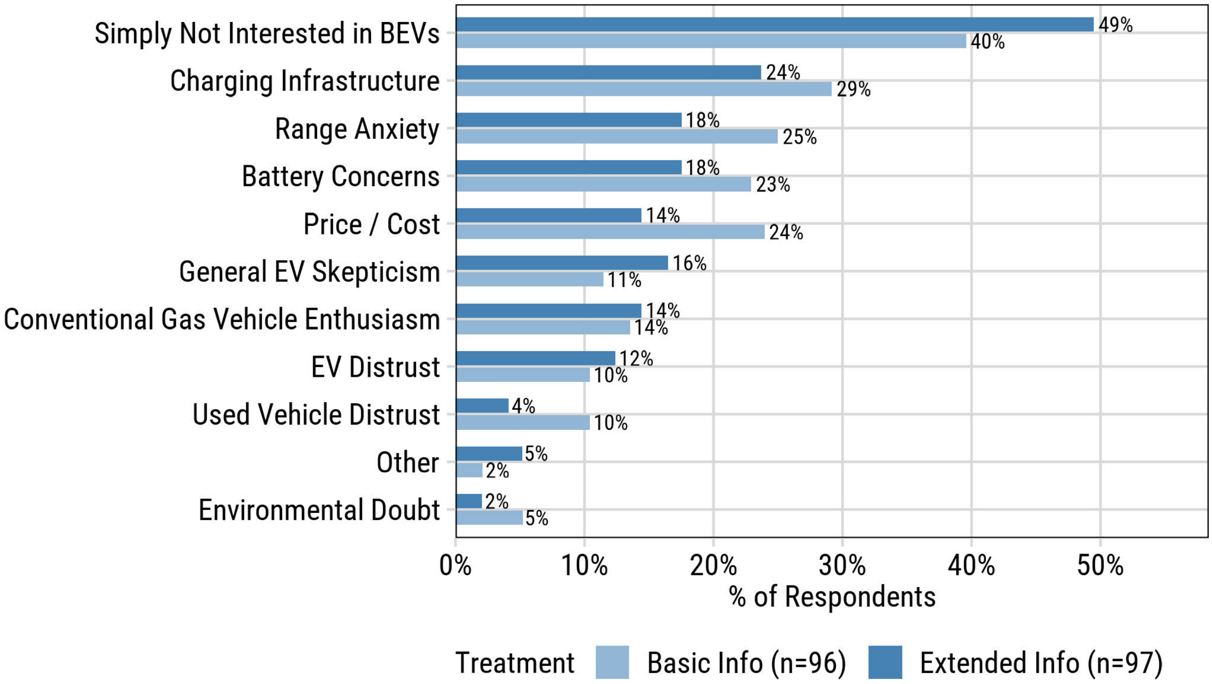

Figure 5: Self-reported opt-out reasons by information treatment group.

6. Conclusion

The used BEV market is expanding rapidly, yet persistent information asymmetry around battery health condition continues to suppress consumer confidence and limit market efficiency. This paper provides the first WTP estimates for battery health information in the U.S. used BEV market, based on a national DCE with 2,916 respondents and a jointly estimated MXL and six-class LCCM.

At the aggregate level, consumers value electric range, penalize mileage and battery degradation, and discount refurbishment history. The MXL estimates confirm that preference heterogeneity is substantial. The LCCM reveals that consumers be organized into six segments that differ markedly in their sensitivity to battery quality signals, their price responsiveness, and their market engagement. **Class 1** (8.7%) is defined by systematic opt-out behavior and represents consumers who are largely outside the addressable used BEV market, with low EV knowledge, high range anxiety, and predominantly ICEV households. **Class 2** (22.7%), the largest class, is characterized by the strongest and most uniform range loss aversion and meaningful disutility for both refurbishment types; these well-informed consumers are effectively purchasing battery reliability rather than range per se. **Class 3** (16.6%) is most EV-engaged and places an exceptionally high premium on crossing a minimum viable range threshold and is the most EV-engaged segment. **Class 5** (20.8%) attends simultaneously to range adequacy, degradation trajectory, and refurbishment condition, and opts out selectively when no presented vehicle meets its standards across all dimensions at once. **Class 6** (12.8%) is the most price-sensitive class and is composed of budget-constrained buyers who are strongly committed to making a vehicle choice but largely indifferent to battery quality signals. **Class 4** (18.5%) is best interpreted cautiously: its near-zero price sensitivity inflates calculated WTP values, and the class likely encompasses a mix of information-treatment-affected respondents, experienced EV adopters, and attribute-disengaged respondents who share a common pattern of large refurbishment disutility alongside imprecise estimates for most other attributes.

These findings carry a direct message for dealers and OEMs. Vehicles with documented performance indicators (mileage, range) and battery health information (degradation rate and refurbished history)represent a material pricing premium opportunity with the majority of the market. It is important to consider the development of standardized battery health certification, analogous in structure and function to vehicle history reports. A certification program that provides independently verified SOH readings, degradation history, and remaining warranty coverage could allow dealers to differentiate inventory and command measurable price premiums from the battery-quality-sensitive majority. Critically, cell and pack replacement are penalized at similar magnitudes despite their technical differences, which indicates that consumers do not currently distinguish between high-quality refurbishment and simple repair. This represents a market opportunity for third-party certification services that validate refurbishment quality and communicate it in terms consumers understand.

The information treatment analysis adds an important qualification: providing extended battery and warranty information does not reduce overall market disengagement, but it redirects the type of concern driving opt-out behavior from infrastructure barriers toward battery-specific anxieties. This finding suggests that disclosure alone is insufficient to build purchase confidence. Credible assurance mechanisms — third-party SOH certification, warranty-backed condition guarantees, and transparent degradation forecasts — are needed to give disclosed battery information a positive commercial meaning rather than surfacing a risk without resolution.

Several limitations should be noted. The study relies on stated preferences from a DCE, which may not fully replicate the decision-making dynamics of real vehicle purchases, particularly for consumers with limited prior EV experience. The sample is restricted to individuals who express an intention to purchase a used vehicle within two years, which excludes passive or uninterested populations. The LCCM assumes a finite number of discrete classes; unobserved within-class heterogeneity remains. Future research should examine how these preferences evolve as used BEV supply increases and battery health reporting becomes more standardized, and whether incentive programs aimed at improving information quality in the secondary market translate into measurable shifts in demand.

CRediT authorship contribution statement

Xiatian Iogansen: Conceptualization, Data curation, Methodology, Software, Formal analysis, Writing - Original Draft, Writing - Review & Editing. **Zain hoda:** Conceptualization, Data curation, Methodology, Writing - Review & Editing. **Christina Gore:** Conceptualization, Data curation, Methodology, Investigation, Writing - Original draft preparation, Writing - Review & Editing, Supervision, Project administration, Funding acquisition. **Joshua D. Kneifel:** Conceptualization, Data curation, Methodology, Investigation, Writing - Review & Editing, Supervision, Project administration, Funding acquisition. **Sindhu Ranganath:** Data curation, Writing - Original Draft, Writing - Review & Editing. **John P.Helveston:** Conceptualization, Methodology, Writing - Review & Editing, Supervision.

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